Algebra Essentials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

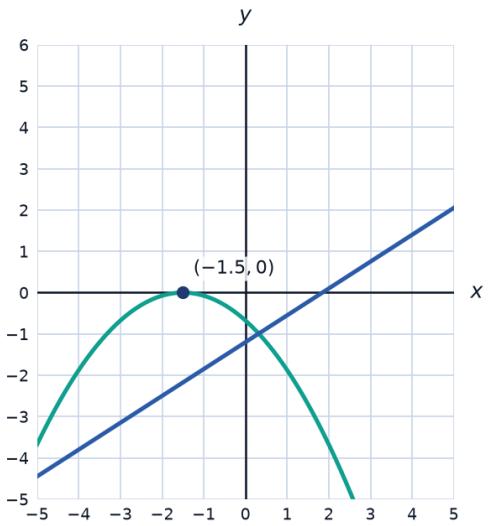
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating expressions

If $x = 4$, evaluate $3x + 7$.

2 [Foundational] • Target: Evaluating expressions

If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.



3 [Foundational] • Target: Evaluating expressions

If $x = -3$, evaluate $2x^2 + 5x$.

4 [Foundational] • Target: Evaluating expressions

If $p = 6$ and $q = -2$, evaluate $\frac{p - q}{q + 6}$.

5 [Foundational] • Target: Solving linear equations

Solve $2x + 5 = 17$.

6 [Foundational] • Target: Solving linear equations

Solve $5x - 8 = 27$.

7 [Foundational] • Target: Solving linear equations

Solve $3(x - 4) = 21$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Solving linear equations

Solve $\frac{x}{4} + 3 = 10$.

9 [Intermediate] • Target: Solving linear equations

Solve $7x + 3 = 4x + 18$.

10 [Intermediate] • Target: Expanding and simplifying

Expand $(x + 3)(x + 5)$.

11 [Intermediate] • Target: Expanding and simplifying

Expand $(x - 4)(x + 6)$.

12 [Intermediate] • Target: Expanding and simplifying

Expand $(2x + 1)(x - 3)$.

13 [Intermediate] • Target: Expanding and simplifying

Simplify $3(2x - 5) - 2(x - 4)$.

14 [Intermediate] • Target: Factoring

Factor $x^2 + 9x + 20$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Factoring • Cross-Lesson: 4.01 Geometry & Measurement

Factor $x^2 - 2x - 15$.

16 [Challenging] • Target: Factoring • Cross-Lesson: 5.01 Quantitative Comparisons

Factor $x^2 - 49$.

17 [Challenging] • Target: Factoring • Cross-Lesson: 6.01 Sequences & Patterns

Factor $6x + 15$.

18 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 7.01 Data Analysis & Probability

Solve $x^2 + 5x + 6 = 0$.

19 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 8.01 Word Problems & Reasoning

Solve $x^2 - 7x + 12 = 0$.

20 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 1.01 Arithmetic & Number Sense

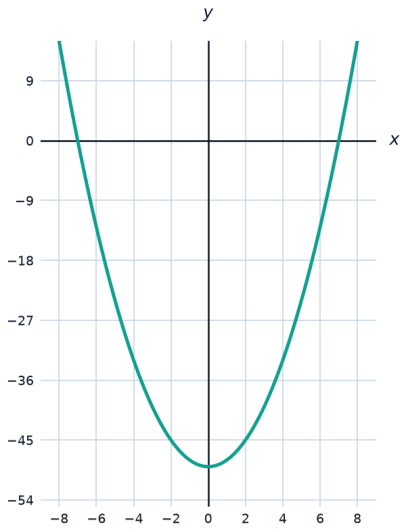
Solve $x^2 - 16 = 0$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving quadratic equations

The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x-axis)?



a. x = 7 only

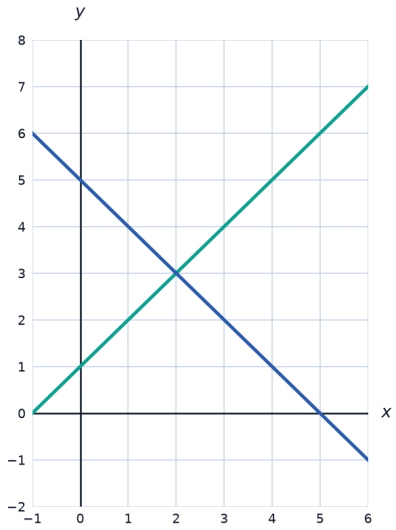
b. x = 49

c. x = 7 or x = -7

d. x = -7 only

22 [Foundational] • Target: Systems of two equations

The graph shows two lines. What is their point of intersection?



a. (1, 4)

b. (0, 5)

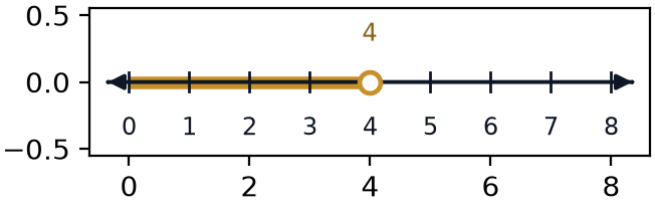
c. (2, 3)

d. (3, 2)

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23 [Foundational] • Target: Inequalities

The number line shows the solution set of an inequality. Which inequality matches?



a. $x \geq 4$

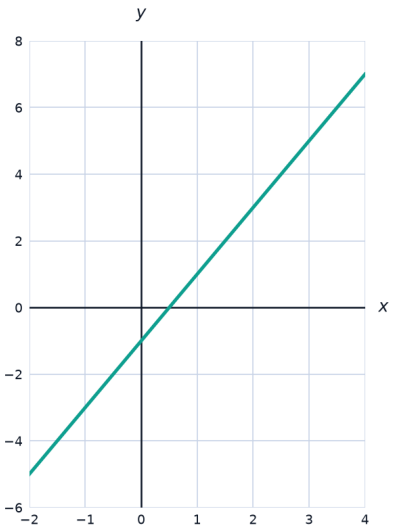
b. $x \leq 4$

c. $x > 4$

d. $x < 4$

24 [Foundational] • Target: Reading function graphs

The graph shows a line. What is its slope?



a. $\frac{1}{2}$

b. 2

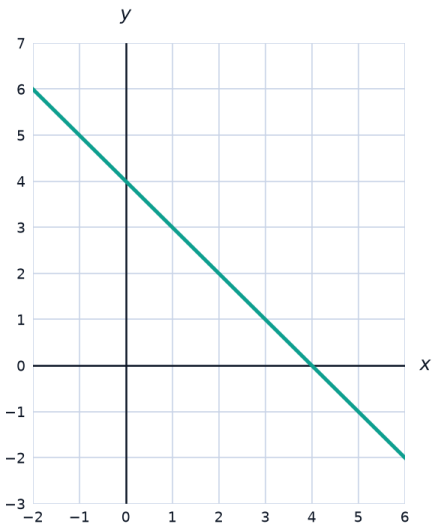
c. -2

d. -1

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25 [Foundational] • Target: Reading function graphs

What is the y-intercept of the line shown?



a. 4

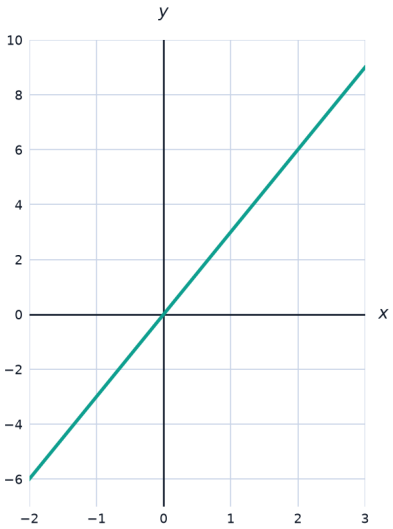
b. 0

c. -4

d. 1

26 [Foundational] • Target: Reading function graphs

What is the equation of the line shown?



a. $y = 3x + 1$

b. $y = x + 3$

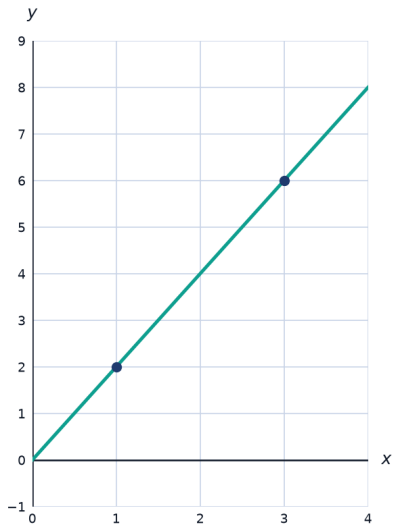
c. $y = \frac{1}{3}x$

d. $y = 3x$

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27 [Foundational] • Target: Reading function graphs

Find the slope of the line through the two marked points.



a. 2

b. $\frac{1}{2}$

c. -2

d. 4

28 [Foundational] • Target: Substitution into formulas

The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w.

a. 8

b. 6

c. 384

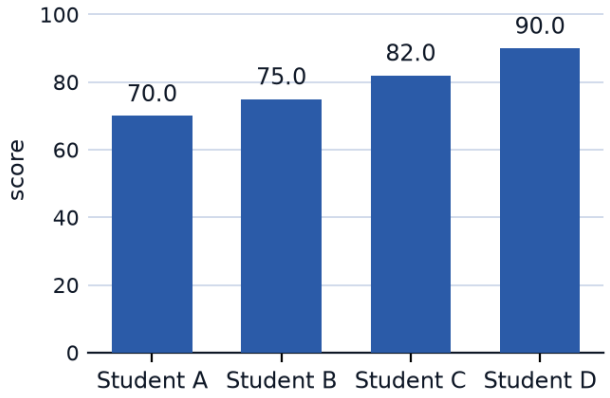
d. 40

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Word problems

The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?



a. Student B

b. Student D

c. Student C

d. Student A

30 [Intermediate] • Target: Evaluating expressions

If $x = 6$, evaluate $4x - 9$.

a. 33

b. 24

c. -15

d. 15

31 [Intermediate] • Target: Evaluating expressions

If $a = -2$ and $b = 3$, evaluate $a^2 + 2ab$.

a. -16

b. 8

c. -4

d. -8

32 [Intermediate] • Target: Evaluating expressions

If $x = 5$ and $y = -1$, evaluate $(x + y)^2 - xy$.

a. 16

b. 11

c. 21

d. -5

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33 [Intermediate] • Target: Evaluating expressions

If $p = 3$, evaluate $2p^3 - 5p$.

a. 49

b. 39

c. 33

d. 69

34 [Intermediate] • Target: Solving linear equations

Solve $4(x + 2) = 3x + 15$.

a. $x = 1$

b. $x = 23$

c. $x = -7$

d. $x = 7$

35 [Intermediate] • Target: Solving linear equations

Solve $\frac{2x - 1}{3} = 5$.

a. $x = 4$

b. $x = 7$

c. $x = 16$

d. $x = 8$

36 [Intermediate] • Target: Solving linear equations

Solve $6 - 2x = 3x - 9$.

a. $x = -3$

b. $x = 3$

c. $x = 15$

d. $x = 1$

37 [Intermediate] • Target: Expanding and simplifying

Expand $(x + 7)^2$.

a. $x^2 + 7x + 49$

b. $x^2 + 14x + 14$

c. $x^2 + 14x + 49$

d. $x^2 + 49$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Expanding and simplifying • Cross-Lesson: 7.01 Data Analysis & Probability

Expand $(x-5)^2$.

a. $x^2+10x+25$

b. $x^2-5x+25$

c. $x^2-10x+25$

d. x^2-25

39 [Challenging] • Target: Expanding and simplifying • Cross-Lesson: 8.01 Word Problems & Reasoning

Expand $(3x-2)(3x+2)$.

a. $9x^2+4$

b. $9x^2-12x-4$

c. $9x^2-4$

d. $6x^2-4$

40 [Challenging] • Target: Expanding and simplifying • Cross-Lesson: 1.01 Arithmetic & Number Sense

Simplify $2(x+3)-3(x-1)$.

a. $5x+9$

b. $-x+9$

c. $x+9$

d. $-x+3$

41 [Challenging] • Target: Factoring • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Factor $3x^2+12x$.

a. $3(x^2+4x)$

b. $x(3x+12)$

c. $3x(x+12)$

d. $3x(x+4)$

42 [Challenging] • Target: Factoring • Cross-Lesson: 4.01 Geometry & Measurement

Factor $x^2-11x+30$.

a. $(x-5)(x-6)$

b. $(x-2)(x-15)$

c. $(x+5)(x+6)$

d. $(x-3)(x-10)$

43 [Challenging] • Target: Factoring • Cross-Lesson: 5.01 Quantitative Comparisons

Factor $x^2+3x-18$.

a. $(x+2)(x-9)$

b. $(x-6)(x+3)$

c. $(x+9)(x-2)$

d. $(x+6)(x-3)$

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44 [Challenging] • Target: Solving quadratic equations • Cross-Lesson: 6.01 Sequences & Patterns

Solve $2x^2 - 8x = 0$.

a. x = 8

b. x = 2 or x = 4

c. x = 0 or x = 4

d. x = 0 or x = -4

45 [Challenging] • Target: Systems of two equations • Cross-Lesson: 7.01 Data Analysis & Probability

Solve the system: $x + 2y = 8$, $x - y = 2$.

a. x = 2, y = 3

b. x = 6, y = 1

c. x = 4, y = 2

d. x = 8, y = 0

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